6 A small coil of wire X is placed inside a solenoid such that the centre of coil X is along the central axis of the solenoid, as shown in Fig. 6.1. The solenoid has 2500 turns per metre. Coil X has 100 turns and a cross-sectional area 7.4×10^{-5} m^2.

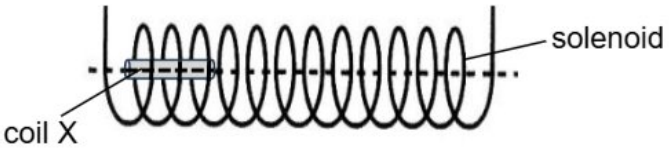


Fig. 6.1

An alternating power supply is now connected to the solenoid. The alternating current I in the solenoid varies with time t according to the following:

$$I = 2.6 \sin (6\pi t)$$

where I is in A and t is in s.

(a) Explain why an e.m.f. is induced in coil X.

.....

.....

.....

..... [2]

(b) Determine the maximum magnetic flux density B_{max} due to the alternating current in the solenoid.

B_{max} = ... T [1]

(c) Calculate the maximum magnetic flux linkage Φ_{max} in coil X.

Φ_{max} = ... Wb-turns [2]

(d) Hence or otherwise, calculate the maximum e.m.f. induced in coil X.

maximum e.m.f. = ... V [2]

[Total:7]